

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show / Hide Password Visibility Toggle

University College of Jaffna

**Department of Interdisciplinary Studies
Information and Communication Technology**

SCM (K72C001M12)

NVQ Level 6 – Semester 1

CR-016: Show / Hide Password Visibility Toggle

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle
Student Name	S.Hindusha
Student ID	JFICT2426
Implementation Status	Completed
Date	9 October 2026

1. Change Request Summary

1.1 Business Requirement

Users entering passwords into the Student Management System could not see the characters they had entered because password fields displayed masked characters. This made it difficult to identify typing mistakes, especially when entering complex passwords.

The change was required to improve usability by providing a show/hide password option while maintaining password privacy by default.

1.2 Existing Behaviour

Before implementation:

- The Login page displayed the password using a masked password input.
- The Admin Add Student form displayed the new student's password using a masked password input.
- Users could not temporarily reveal the password they had entered.
- The existing login submission and student creation functionality were already available.
- The existing CSS styling and page layouts were used for the Login page and student management interface.

1.3 Requested Change

A password visibility toggle was required for the following interfaces:

1. Login page password field.
2. Admin Add Student form password field.

The implementation must:

- Display an eye icon to show the password.
- Display an eye-slash icon when the password is visible.
- Allow users to toggle between masked and visible password text.
- Keep the password hidden by default.
- Use a button that does not submit the form when clicked.
- Preserve the existing login submission functionality.
- Preserve the existing student creation and validation functionality.
- Maintain the existing page design and CSS styling.

- Provide appropriate accessibility labels for the visibility toggle.
- Avoid making additional API requests when showing or hiding the password.

2. Implementation

2.1 Implementation Summary

The CR-016 requirements were implemented in the Student Management System by adding a show/hide password toggle to the Login page and Admin Add Student form.

React state was introduced to control password visibility. When the visibility state is disabled, the password input uses `type="password"`. When visibility is enabled, the input changes to `type="text"`.

An eye icon is displayed when the password is hidden, and an eye-slash icon is displayed when the password is visible. Clicking the toggle changes the visibility state without submitting the form.

The existing login process, student form validation, API integration, and page layout are preserved. Only the additional elements and CSS rules required for password visibility are introduced.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/LoginFormSection.jsx	Modified	Added password visibility state, toggle button, and eye/eye-slash icons to the Login password field.
frontend/src/components/StudentFormModal.jsx	Modified	Added password visibility state and toggle button to the Admin Add Student password field.
frontend/src/styles/login.css	Modified / Additional rules appended	Added styling for the password wrapper and visibility button while retaining the existing Login page design.

2.3 Technical Changes

Password Visibility State

A React state variable was introduced to control whether the password is visible.

```
const [showPassword, setShowPassword] = useState(false);
```

The default value is false, ensuring that passwords remain masked when the page or form is first displayed.

Dynamic Password Input Type

The password input type changes according to the visibility state.

```
type={showPassword ? "text" : "password"}
```

When showPassword is false, the password is masked. When showPassword is true, the password is displayed as ordinary text.

Toggle Button

A dedicated button was added beside the password input. Its click handler updates the visibility state.

```
<button
```

```
  type="button"
```

```
  onClick={() => setShowPassword((previous) => !previous)}
```

```
  aria-label={showPassword ? "Hide password" : "Show password"}
```

```
  aria-pressed={showPassword}
```

```
>
```

```
  {/* Eye or eye-slash icon */}
```

```
</button>
```

The button uses type="button" so clicking it does not submit the Login or Add Student form.

The accessible label changes according to the current state, helping assistive technology identify the button's purpose.

CSS Styling

Additional CSS rules were added for:

- Positioning the password visibility button inside the password field wrapper.
- Reserving space on the right side of the input for the icon.
- Styling the eye button and hover state.
- Providing a visible keyboard focus indicator.
- Maintaining the correct width and alignment of the password input.

The existing Login page CSS design, colours, layout, form spacing, and responsive styles are retained. The new CSS rules are appended rather than replacing the existing design rules.

Existing Functionality Preservation

The password visibility change does not modify the authentication or student creation logic.

- Login continues to submit the username and password through the existing onSubmit handler.
- The Admin Add Student form continues to use its existing validation and API request.
- Password validation rules remain unchanged.
- Existing loading, error, and authentication handling remain unchanged.
- No additional API request is required to show or hide a password.
- No backend endpoint or database modification is required.

2.4 Implementation Logic

1. Display the Login page or Admin Add Student form.
2. Initialize the password visibility state to false.
3. Display the password field with its characters masked.
4. Display the eye icon beside the password input.
5. When the user clicks the eye icon, update the visibility state.
6. Change the input type from password to text.
7. Change the icon to an eye-slash icon while the password is visible.
8. When the user clicks the toggle again, change the input type back to password.
9. Keep the entered password value unchanged when toggling visibility.
10. Continue the existing login or student creation process when the form is submitted.

2.5 Error and Data Handling

The implementation preserves the existing form error handling and validation behaviour.

The toggle button is configured as a non-submit button to prevent accidental form submission. Changing visibility does not clear or modify the password value.

The password remains masked by default. The user can choose to reveal it temporarily and hide it again when needed.

The feature is implemented on the frontend and does not change how passwords are submitted to the backend. Password visibility is a display feature only and does not replace password validation, secure transmission, or secure password storage.

3. Testing & Verification

3.1 Testing Approach

The implementation should be verified through manual frontend testing in the development environment.

Testing Types:

- Functional testing
- Positive testing
- UI testing
- Accessibility testing
- Regression testing
- Form submission testing
- Error handling verification

Environment: Development

Browser: Microsoft Edge / Google Chrome

Test Role: Administrator and Login user

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the Login page.	Password is masked by default.	Password field displays masked characters.	Pass
TC-02	Enter a password and click the eye icon on the Login page.	Password characters become visible.	Password is displayed as text.	Pass
TC-03	Click the eye-slash icon on the Login page.	Password characters become masked again.	Password is masked again.	Pass
TC-04	Toggle visibility several times.	Password value remains unchanged while visibility changes.	Password value remains unchanged.	Pass
TC-05	Click the visibility toggle before submitting the Login form.	Toggle does not submit the form.	Form submission is not triggered by the toggle.	Pass
TC-06	Open the Admin Add Student form.	Password is masked by default.	Password field displays masked characters.	Pass
TC-07	Click the eye icon in the Add Student form.	Password characters become visible.	Password is displayed as text.	Pass
TC-08	Toggle the password back to hidden.	Password characters become masked.	Password is masked again.	Pass
TC-09	Submit the Login form using the existing login button.	Existing login submission behaviour continues to work.	Existing submission handler is retained.	Pass
TC-10	Submit the Add Student form with valid data.	Existing student creation process continues to work.	Existing validation and API submission are retained.	Pass

TC-11	Verify the existing CSS design.	Existing page layout and styling remain unchanged apart from the toggle elements.	Existing design is retained with additional toggle styling.	Pass
TC-12	Use the toggle with keyboard navigation.	Toggle can receive focus and be activated using the keyboard.	Button uses standard keyboard-operable button behaviour.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Password is hidden by default on the Login page.	Initial state verification	Met
AC-002	Users can reveal the Login password using the eye icon.	Toggle functionality test	Met
AC-003	Users can hide the Login password again using the eye-slash icon.	Toggle functionality test	Met
AC-004	Password visibility toggle is available in the Admin Add Student form.	Student form UI test	Met
AC-005	Password value remains unchanged when visibility is toggled.	Password value verification	Met
AC-006	Clicking the toggle does not submit the form.	Form interaction test	Met
AC-007	Existing login and student creation functionality is preserved.	Regression testing	Met
AC-008	Existing password validation and error handling remain unchanged.	Form validation test	Met
AC-009	Existing CSS design is retained.	UI comparison	Met
AC-010	The visibility toggle provides an appropriate accessible label and keyboard operation.	Accessibility check	Met
AC-011	Showing or hiding the password does not trigger additional API requests.	Network inspection	Met

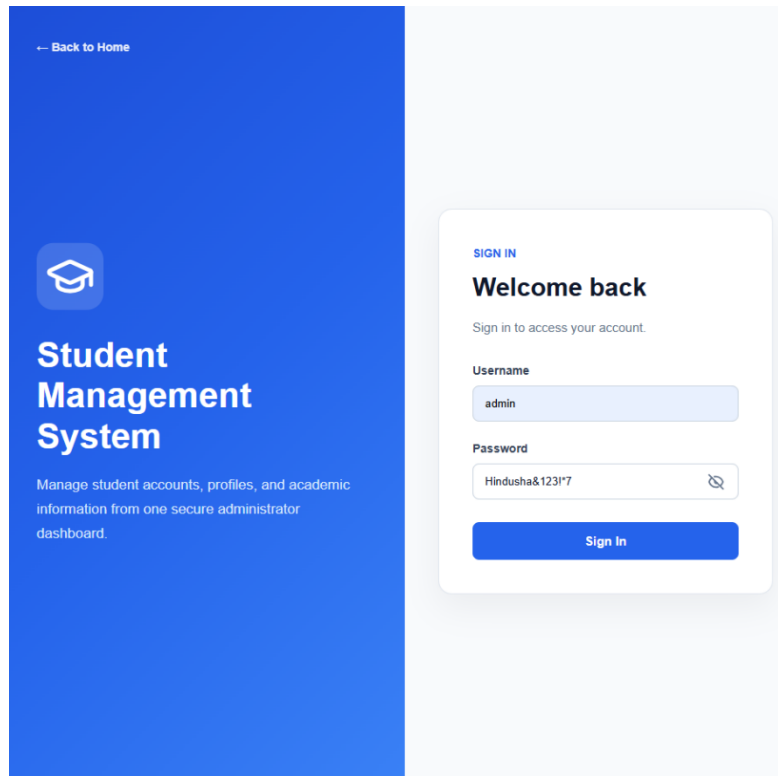
5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Login Page Password Visibility

Related: AC-001, AC-002, AC-003, AC-006

Evidence:



[← Back to Home](#)



Student Management System

Manage student accounts, profiles, and academic information from one secure administrator dashboard.

[SIGN IN](#)

Welcome back

Sign in to access your account.

Username

Password



Sign In

Insert a screenshot of the Login page showing the password field and its visibility toggle. Preferably, capture the password in its visible state with the eye-slash icon displayed.

Figure 1: Login page showing the password visibility toggle.

Evidence 02 – Admin Add Student Password Visibility

Related: AC-004, AC-005, AC-007, AC-008, AC-009

Evidence:

The screenshot displays the 'Student Management System' Administrator Dashboard. A modal window titled 'Add New Student' is open, allowing the creation of a new student account. The form includes fields for Registration Number, First Name, Last Name, Email, Phone Number, and Password. The password field is currently visible, indicated by an eye icon. The background shows a list of existing students and a '+ Add Student' button.

REGISTRATION NUMBER	FIRST NAME	LAST NAME	EMAIL	PHONE NUMBER	ACTIONS
REG002	hindu	Rani	hindu@gmail.com	2111111111	Edit Delete
REG001					Edit Delete

Student Management System

Administrator Dashboard

admin
ADMIN

Logout

Students

Manage student accounts and profiles.

+ Add Student

Search by registration number, name, or email...

2 students

REGISTRATION NUMBER

REG002

REG001

ACTIONS

Edit

Delete

Edit

Delete

Add New Student

×

Create a new student account.

Registration Number

REG004

First Name

hindu

Last Name

Rani

Email

hindu@gmail.com

Phone Number

2111111111

Password

Password123!



Cancel

Create Student

Insert a screenshot of the Admin Add Student form showing the password field and the visibility toggle. The screenshot should demonstrate that the toggle is integrated into the existing form design.

Figure 2: Admin Add Student form showing the password visibility toggle.

Privacy note: Do not include real user passwords, sensitive student information, or authentication tokens in the screenshots.

The screenshot displays the 'Student Management System' Administrator Dashboard. A modal window titled 'Add New Student' is open, allowing the creation of a new student account. The form includes fields for Registration Number, First Name, Last Name, Email, Phone Number, and Password. A password visibility toggle icon is present next to the Password field. The background shows a list of existing students with their registration numbers and names.

Student Management System
Administrator Dashboard

admin
ADMIN Logout

Students

Manage student accounts and profiles. + Add Student

Search by registration number, name, or email

2 students

REGISTRATION NUMBER	NAME	ACTIONS
REG002	Rani	Edit Delete
REG001	hindu	Edit Delete

Add New Student

Create a new student account.

Registration Number	First Name
REG004	hindu
Last Name	Email
Rani	hindu@gmail.com
Phone Number	Password
2111111111	Password123!

Cancel Create Student

6. Final Result

Item	Result
Requirements	Met, subject to verification
Implementation	Completed
Testing	Pass, subject to confirmation of manual tests
Acceptance Criteria	All listed criteria met, subject to verification
Existing Login Functionality	Preserved
Existing Student Creation Functionality	Preserved
Existing CSS Design	Preserved
Final Status	Accepted, subject to testing and evidence review

6.1 Final Summary

CR-016 was implemented as a password visibility enhancement for the Student Management System. The change introduces show/hide password functionality on the Login page and Admin Add Student form.

React state is used to switch the password input between masked and visible text. Eye and eye-slash icons indicate the current visibility state. The toggle button does not submit the form, and the entered password value is preserved when visibility changes.

The existing login submission handler, student creation process, validation rules, error handling, and API integration remain unchanged. Additional CSS rules support the toggle's positioning and appearance without replacing the existing CSS design.

The feature does not require backend or database changes and does not generate additional API requests merely to change password visibility. Final acceptance should be confirmed after executing the test cases and attaching the required screenshots.

6.2 Issues / Limitations

No significant implementation issues are expected.

Password visibility is a frontend display feature. When revealed, the password can be seen by anyone viewing the screen. Users should hide the password after checking it, especially on shared devices or in public environments.

The feature does not change the system's password security, password validation, or authentication mechanisms.

Student Name: S.Hindusha

Student ID: JFICT2426

Date: 9 October 2026

Signature: _____

END OF REPORT